

PAHADICART (पहाड़ीकार्ट) MASTER ARCHITECTURE & OPERATIONS USER MANUAL

Complete End-to-End System Flow, Terrain Dispatch Engine, Triad Operations, and Step-by-Step User Manual for Admins, Merchants, Riders & Customers

PLATFORM NAME

PahadiCart ("Aapka Apna Pahadi Bazaar")

OPERATIONAL HUBS

Solan (1,502m) • Shimla (2,206m) •
Dharamshala (1,457m)

INTEGRATED PORTALS

Super Hub, Admin Tower, Customer App,
Merchant POS, Rider Cockpit

ADVANCED MODULES

WhatsApp Commerce Bot, Batched Staircase
Route, 58mm Thermal KOT

TECH ARCHITECTURE

100% Zero-Cost Open Source (Vanilla JS, Web
Audio, Leaflet, PWA)

TARGET AUDIENCE

Developers, Operations Executives,
Merchants, Riders & Support

PahadiCart Hyperlocal Ecosystem

Empowering Himalayan High-Street Vyapar Mandal Commerce with 2-Hour
Delivery.

Edition: September 2026

Status: **Live Production**
(Vercel)

1. Executive Summary & Complete System Architecture

PahadiCart (पहाड़ीकार्ट) is Himachal Pradesh's first purpose-built 2-hour hyperlocal marketplace and mountain delivery logistics engine. Unlike metropolitan quick-commerce that burns cash on 10-minute dark stores, PahadiCart unites existing ancestral high-street shops (**Himachal Vyapar Mandal**) with a sustainable, elevation-aware delivery fleet.

Zero-Cost Open Source Stack

₹0 APIs

Built entirely on standard browser APIs (Web Audio API for temple bells, Web SpeechSynthesis for Hindi voice alerts, Leaflet OSM for maps, Service Worker for PWA, and Node.js).

Mountain Terrain Engine

Elevation-Aware

Dynamically calculates base delivery fees (₹25), staircase climb fees (₹25), steep slope elevation fees (₹15), and weather surge buffers for rain (+₹15) and snow/landslide curfews.

Multi-Modal Logistics

Vehicle + Foot

Combines mountain scooters (Hero Splendor, EV scooters) with registered walking runners for pedestrian mall roads (Shimla Ridge & Mall Road, McLeod Ganj Square).

1.1 The 5 Integrated Portals at a Glance

Portal Name	Live Production URL	Primary Users	Key Functions
 Super Hub Platform Launcher	https://pahadicart.vercel.app/	Founders, Investors, Platform Admins	One-stop ecosystem dashboard, 1-Click Lifecycle Simulator trigger, global portal switcher.
 Super Admin Command Tower	https://pahadicart.vercel.app/admin/	Operations Team, City Managers	Live Triad Tracker, Rider GPS fleet radar, financial P&L ledger, multi-town weather surge controls.
 Customer PWA Web App	https://pahadicart.vercel.app/customer/	Residents, Tourists, Homestays	Multi-town catalog browsing, WhatsApp Hinglish bot, staircase address profiling, offline tunnel mode.
 Vyapar Mandal Merchant Shop	https://pahadicart.vercel.app/merchant/	Kirana owners, Bakers, Dhabas, Chemists	Ghar Ki Ghanti audio chime, Kanban order dispatch board, 58mm POS thermal receipt & KOT printer.
 Pahadi Rider Mobile Cockpit	https://pahadicart.vercel.app/rider/	Scooter Riders & Walking Runners	Glove-friendly touch UI, batched route multi-order optimizer, staircase instructions, doorstep OTP verification.

2. End-to-End Mountain Order Lifecycle (5-Stage Triad Flow)

Every transaction on PahadiCart executes across a unified real-time event pipeline (**PahadiEventBus**) connecting the Customer, Merchant, Rider, and Admin simultaneously without data loss.

[Customer App / WhatsApp Bot] | 1. Order Placed (Items + Staircase Address + Elevation Fee) ↓ [PahadiEventBus & Dispatch Engine] → Triggers "Ghar Ki Ghanti" Web Audio Chime | 2. Proximity Shopkeeper Assignment ↓ [Merchant Portal] → Accepts & Prints 58mm KOT Slip (Prepares Package) | 3. Auto-Dispatches to Nearest Available Rider ↓ [Rider Cockpit] → Radar Ping → Accepts & Navigates Batched Mountain Trail | 4. Staircase Climb & Doorstep Handover ↓ [Customer Handshake] → Secret 4-Digit OTP Verified on Doorstep (COD / UPI Collected) | 5. Immediate Financial Settlement ↓ [Admin Command Tower] → Live GMV, Platform Commission, Rider Payout & 1% TCS Credited

Detailed 5-Stage Step Breakdown

- STAGE 1 • CUSTOMER ORDERING & ADDRESS PROFILING

Order Placement via PWA App or WhatsApp Conversational Bot

The customer selects items from their local town catalog (Solan, Shimla, or Dharamshala). During checkout, they specify hill-specific landmark details:

 - **Staircase Guidance:** e.g., *"Descend 35 stone steps from main road level, green gate"*.
 - **Dynamic Fee Computation:** Base Delivery (₹25) + Staircase Doorstep Fee (₹25) + Mountain Elevation Surge (₹15) + Dynamic Weather Surge (if rain/snow active).
 - Alternatively, the customer types in Hindi on WhatsApp bot: *"Bhaiya 1kg seb aur 2 Siddu bhej do Shamti, 42 seedhiyo ke upar"* — the NLP bot extracts items and location automatically.
- STAGE 2 • NEAREST SHOPKEEPER DISPATCH & "GHAR KI GHANTI"

Merchant Alert, Audio Chime & KOT Generation

The system locates the closest merchant within 2.5km having stock:

 - The merchant portal sounds a loud brass temple bell chime ("**Ghar Ki Ghanti**") via Web Audio API.
 - If ordered via WhatsApp, an automated Hindi voice IVR calls the shopkeeper: *"Namaskar Sharma Ji! Naya WhatsApp order aaya hai..."*
 - The merchant reviews items and clicks "📄 **KOT Slip**" to generate a 58mm/80mm receipt with customer staircase notes and kitchen item list.
 - Merchant moves status from *Placed* to *Preparing*.
- STAGE 3 • RIDER PROXIMITY RADAR & MULTI-ORDER BATCHING

Ascending Staircase Routing & Trail Climb

The dispatch engine broadcasts the order to active riders in that town:

 - Nearest rider with cash-in-hand < ₹2,500 receives an incoming radar ping with payout (₹85), hill climb (+140m), and ETA.
 - If multiple orders are pending in the same valley pocket, the **Batched Route Optimizer** clusters stops in an ascending elevation sequence (carrying heaviest loads at the bottom and climbing upward), reducing hill fatigue by ~38%.
 - Rider clicks *Accept*, rides to the store, collects the package, and heads toward the delivery point.

STAGE 4 • DOORSTEP STAIRCASE HANDOVER & OTP VERIFICATION

Proof of Delivery & Cash Collection Safeguard

Rider reaches the hill coordinate, navigates the stone stairs using customer notes, and meets the customer:

- Customer shows their secret 4-digit OTP (e.g. 2460) shown on their screen or WhatsApp chat.
- Rider enters OTP into their glove-friendly input box. If incorrect, system blocks completion.
- If COD, rider collects cash (displayed prominently) and floating wallet balance updates. If UPI, customer scans the QR printed on the thermal receipt.

STAGE 5 • INSTANT SETTLEMENT & FINANCIAL RUPEE WATERFALL

Automated Platform Margin & Government TCS Deductions

Upon OTP verification, the transaction instantly settles across the Super Admin Tower:

- **Gross Merchandise Value (GMV):** Credited to platform daily gross.
- **Platform Net Margin:** Retained automatically (6% to 18% category commission + platform fee).
- **Rider Payout:** Base fee + KM rate + staircase fee + tips credited directly to rider daily wallet.
- **Section 194–O 1% TCS:** Automatically computed, deducted, and logged for merchant Form 26AS income tax filing.

3. Operations User Manual: Super Admin Command Tower

The **Super Admin Command Tower** (/admin/) is the central nervous system for platform founders, city operations managers, and customer support leads.

3.1 How to Monitor Operations in Real-Time

1. Triad Tracker (Customer + Rider + Shopkeeper)

Live
View

Located in the **Live Triad Tracker** tab. Displays every active order as an integrated 3-party card:

- Shows Customer Name, phone, and exact staircase descent instructions.
- Shows Merchant Shop Name, Vyapar Mandal reg, and packing status.
- Shows Assigned Rider, vehicle plate, battery %, and altitude climbed.
- Live timer tracking delivery SLA (Standard 45m vs Weather Buffered).

2. Multi-Town GPS Fleet Radar

Leaflet OSM

Located in the **Fleet Telemetry & Maps** section:

- Interactive OpenStreetMap centered on Solan, Shimla, or Dharamshala.
- Green scooter markers indicate live rider positions and movement.
- Store markers show merchant locations; house markers show delivery pins.
- Clicking any rider displays their current battery, signal, and COD cash-in-hand.

3.2 Step-by-Step Instructions for City Managers

Task A: How to Run the 1-Click Live Lifecycle Simulator

Used for daily system testing, merchant training, and investor demonstrations:

1. Navigate to the top right of the Admin Command Tower and click the green button: " ⚡ 1-Click Simulator".
2. In the modal that appears, select the target town: **Solan (Pine Valley)**, **Shimla (The Ridge)**, or **Dharamshala (McLeod Ganj)**.
3. Click " 🚀 Run Live Mountain Simulation".
4. Watch the 5 live stages automatically execute over 10 seconds:
 - Stage 1: Customer order drafted with realistic staircase drop.
 - Stage 2: Nearest merchant auto-selected & "Ghar Ki Ghanti" audio chime sounds.
 - Stage 3: Nearest rider assigned & elevation hill climb simulated.
 - Stage 4: Doorstep OTP automatically validated.
 - Stage 5: Live revenue and profit added to Admin P&L ledger with an emerald glow.
5. Click *Close* and verify the new order in the Orders Table.

Task B: How to Adjust Dynamic Weather Surges (Rain / Snow / Landslide)

1. In the top header bar, locate the **Dynamic Weather & Surge Control** selector.
2. Select the current weather condition:
 - 🌞 **Clear Skies**: Standard pricing, 45m SLA window.
 - 🌧️ **Monsoon Rain**: Adds +₹15 rain surge to customer bill (100% passed to rider) and adds +20m to delivery ETA buffer.
 - ❄️ **Snow & Landslide Emergency**: Adds +₹35 surge, restricts deliveries to walking runners, and blocks steep two-wheeler slope routes.
3. Changes apply immediately to all customer carts and rider dispatch cards across that town.

Task C: How to Handle Emergency SOS Breakdown Alerts

When a delivery rider experiences a mechanical failure, landslide blockage, or slip:

1. An emergency crimson banner appears at the top of the Admin dashboard with an SOS alert sound.
2. Click the **SOS Alert** card to view the rider's last known GPS coordinate, altitude, and current customer order ID.
3. Click " **Call Rider**" to verify safety.
4. Click "**Re-dispatch Order**" to automatically transfer the package to the second nearest rider without cancelling the customer order.

4. Operations User Manual: Vyapar Mandal Merchant Shop Portal

The **Merchant Shop Portal** (/merchant/) is designed for local Himachal dukandars, halwais, bakers, and chemists. It operates seamlessly on a low-cost Android tablet, desktop PC, or smartphone behind the billing counter.

4.1 Receiving Orders & Kitchen Dispatch

Step 1: Listening for the "Ghar Ki Ghanti" & Voice Call

Jab dukan khuli ho (status:  *Dukan Khuli Hai (Live)*), counter tablet ko hamesha volume ON par rakhein:

- Naya order aate hi screen par chamakdaar yellow/green card aayega aur ek traditional "**Ghar Ki Ghanti**" (**Mandir ki Ghanti**) bajegi.
- Agar order customer ne WhatsApp ke zariye bheja hai, toh speaker par saaf Hindi me aawaz aayegi: "*Namaskar Sharma Ji! Naya WhatsApp order aaya hai...*"
- Order card par order number, customer ka naam, colony, aur samaan ki list dikhegi.

Step 2: Printing the 58mm / 80mm Thermal Receipt & KOT Slip

Counter par kitchen ya packing staff ke liye slip print karna behad aasaan hai:

1. Order card par ya upar header me " **KOT Slip**" button dabayein.
2. Ek pop-up preview khulega jisme official **Himachal Vyapar Mandal** format me slip ban kar aayegi:
 - Dukan ka naam, pata, aur Vyapar Mandal Reg number (e.g. VM-SOL-2024-089).
 - **Staircase Guidance Box**: Isme likha hoga ki customer ka ghar road se kitni seedhiyan upar/neeche hai taaki packing theek se ho (jaise liquid items ko leak-proof pack kiya jaye).
 - Items ki Hindi/English list, quantity, aur price.
 - Customer scan-and-pay ke liye high-contrast **UPI QR Code**.
 - Government 1% Section 194-O TCS tax deduction breakdown.
3. Slip par "**Print Slip Now**" dabayein ya 58mm Bluetooth thermal printer se 1-second me print nikal kar bag par chipka dein.

Step 3: Moving Order on the Kanban Board

- Samaan pack hote hi order card par "**Accept & Pack** (स्वीकार करें)" dabayein.
- Card *Incoming Placed* se *Preparing & Ready* column me chala jayega.
- Rider jaise hi dukan par aakar samaan uthayega, card automatically *Out for Delivery (Rider Picked)* me move ho jayega.

Step 4: Using the Camera Barcode Scanner for Instant Stock Updates

Dukandar counter tablet ya phone ke camera se product packaging ka barcode scan karke 1 second me stock update kar sakta hai:

1. Header me " **Barcode Scanner**" button dabayein.
2. Screen par animated red laser scan frame khulega. Packaging barcode ko green box me align karein.
3. Barcode detect hote hi audio laser beep sunai degi aur screen par " **Barcode Matched! New Stock: +1 unit**" ka message aur Hindi voice bolkar confirm karegi.
4. Desktop testing ke liye modal ke andar pre-set grocery barcodes (Apples, Siddu, Rajma, Tea) par 1-tap karke bhi stock update kiya ja sakta hai.

4.2 Daily Vyapar Mandal Settlement & Tax Ledger

Dukandar ke dashboard ke sabse upar ek daily settlement strip di gayi hai jo transparent accounts maintain karti hai:

Metric Field	What it Shows	Shopkeeper Action / Benefit
Aaj Ke Kul Orders	Aaj dukan se deliver huye kul orders ki sankhya.	Order volume monitoring.
Gross Sales (Bikri)	Aaj ka total sales turnover.	Real-time daily income view.
Platform Commission	Category ke hisaab se 6% to 15% platform cut.	Zero hidden charges. Only agreed Vyapar Mandal rate.
1% TCS Tax Deduction	Income Tax Act Section 194-O ke tahat 1% government tax.	Aapka paisa doobta nahi: Ye amount directly aapke PAN card par Form 26AS me credit hota hai jise ITR me refund le sakte hain.
Net Bank Payout	Aapke bank account me transfer hone wala final amount.	Bank account me daily subah UTR number ke saath jama hota hai.

5. Operations User Manual: Pahadi Rider Mobile Cockpit

The **Pahadi Rider Mobile Cockpit** (/rider/) is custom-designed for local mountain riders using smartphones mounted on motorcycle handlebars or carried in pockets with woolen riding gloves.

5.1 Glove-Friendly Touch & High-Altitude Safety

- **56px Large Hit Targets:** Buttons are extra large so riders never have to take off winter riding gloves.
- **High-Contrast Dark Theme:** Optimized for glaring mountain sun, thick morning fog, and dark hill roads without streetlights.
- **Live Elevation Telemetry:** Shows altitude climbed (+420m), temperature, signal strength, and accumulated mountain climb bonus.

5.2 Step-by-Step Rider Delivery Routine

Step 1: Going Online & Duty Radar

1. Rider app khol kar upar "  **Online (On Duty)** " button verify karein. (Agar chai break par hain, toh 1-click me Offline ho sakte hain).
2. Jaise hi aas-paas ke kisi merchant shop par naya order aayega, rider ke phone me loud alert ping bajega.
3. Radar card par clearly dikhega:
 - **Kamai (Payout):** e.g. ₹85 Payout (Base fee + Elevation bonus + Tips).
 - **Pickup:** Merchant ka naam aur location (e.g. *Sharma Kirana, Mall Road*).
 - **Drop:** Customer ki colony aur staircase profile (e.g. *Shamti, Upper Pine Lane*).
 - **Route Metrics:** 2.4 km doori, +140m climb, 22 min SLA.
4. Bada green button dabayein: "  **Accept Delivery Task (स्वीकार करें)** ".

Step 2: Using the " Batched Route " Staircase Optimizer

Subah ke waqt (Doodh, bread, medicines, sabzi) agar rider ko ek hi pahaad me 2 ya 3 orders deliver karne hain:

1. Header me "  **Batched Route** " button par click karein.
2. Modal khulega jo stops ko **Ascending Staircase Sequence (Neeche se Upar)** me arrange kar deta hai:
 - **Stop 1 (Neeche wala ghar):** Bhaari grocery bag pehle deliver ho jata hai.
 - **Stop 2 & 3 (Upar wali ridge):** Rider halke bag ke saath upar chadhta hai.
3. Screen par dikhega: " ~38% Hill Fatigue Reduced " aur "20 mins Time Saved".
4. Click karein: "  **Start Batched Route Navigation** " — GPS trail seedhe shuru ho jayega.

Step 3: Staircase Navigation & Doorstep OTP Verification

1. Customer ke diye gaye landmark par bike safe jagah park karein.
2. Screen par customer ki  **Staircase Alert** padhein (jaise: "Descend 35 stone steps from road level, green gate").
3. Doorstep par pahunch kar customer se unke phone par dikh raha **4-Digit Secret OTP** poochein.
4. App me OTP daalein (e.g. 2460). Galat OTP daalne par delivery complete nahi hogi.
5. Agar order **COD (Cash on Delivery)** hai: Customer se cash collect karein (screen par exact amount yellow me dikhega).
6. Click karein: "  **Complete Delivery & Add ₹85 to Wallet** ". Badhai ho! Payout turant rider ke daily balance me jud jayega.

Step 4: Emergency SOS Trigger (Landslide / Gaadi Breakdown)

Agar raste me bike puncher ho jaye, rasta band mile, ya landslide ho:

- Screen ke upar red button dabayein: "🚨 **EMERGENCY SOS: Hill Breakdown**".
- Admin Control Tower ko turant rider ke live GPS coordinates [Lat, Lng] ke saath emergency rescue alert chala jayega.
- Support team rider ko phone karegi aur unka order backup rider ko transfer kar degi.

6. Customer Guide, Multi-Town Usage & Troubleshooting FAQ

6.1 Customer Ordering: App vs WhatsApp Bot

Method A: Customer PWA Web App

Visual Catalog

1. Open <https://pahadicart.vercel.app/customer/> on phone or desktop browser.

2. Select town: **Solan (1,502m)**, **Shimla (2,206m)**, or **Dharamshala (1,457m)**.

3. Add local specialties (e.g. Kinnaur Apples, Fresh Siddu Dough, Kangra Tea, Thukpa).

4. Enter doorstep staircase details in address notes.

5. Pay via UPI or select Cash on Delivery (COD).

Method B: WhatsApp

1-Click Hinglish

Conversational Bot

1. Tap the green floating **WhatsApp Icon** on the bottom right.

2. Type naturally in Hindi: *"Bhaiya 1kg seb aur 2 Siddu bhej do Shamti, 42 seedhiyo ke upar"*.

3. Bot automatically extracts items, colony, adds staircase fee, and confirms the order with OTP.

4. Merchant counter par "Ghar Ki Ghamti" bajti hai aur Hindi call chali jaati hai.

6.2 "Bol Kar Order Karein" (Voice-to-Cart Engine)

Buzurg ya gaadi chalate waqt typing se bachne ke liye:

1. Customer App header me " बोलकर ऑर्डर करें" button dabayein.

2. Screen par pulsing green microphone icon khulega. Hindi ya Pahadi bolchaal me bolein: *"Bhaiya 2 Siddu aur 1kg seb bhej do Shamti"*

3. System aawaz ko recognize karke seedhe cart me items add kar dega aur Hindi me aawaz aayegi: *"Shandar! Aapke cart mein 2 items add kar diye gaye hain."*

6.3 Live Animated GPS Mountain Radar Tracking

Order place hone ke baad live delivery scooter ko pahaadon par move karta hua dekhein:

- Header me " **Live Rider GPS**" button dabayein.

• Leaflet map par rider ka scooter actual hairpin turns aur hill curves par smoothly move karta hua dikhega.

• Upar telemetry HUD me live altitude climb (1, 502m → 1, 640m), speed (24 km/h), bachi hui doori (1.6 km) aur ETA live countdown hoga.

• Rider card par " **Call Rider**" button se masked call ki ja sakti hai.

6.4 Tunnel Mode (Deep Valley Offline Queuing)

Pahadon me jab 4G/5G signal chala jata hai (road tunnels ya deep gorges me):

- Customer app par header me " **Tunnel Mode (Offline)**" indicator toggle ho jata hai.

• Customer offline rehkar bhi samaan cart me daal kar order place kar sakta hai.

• Order device ke secure local queue me save rehta hai (status: `queued_offline`).

• Jaise hi vehicle tunnel se bahar aayegi aur signal aayega, order automatically cloud server par sync ho jayega!

6.3 Comprehensive Troubleshooting & Operations FAQ

Issue / Question	Underlying Cause	Step-by-Step Resolution
"Ghar Ki Ghamti" sound nahi aa rahi?	Modern browsers block auto-playing audio without first user interaction.	Merchant dashboard par header me " Test Ghar Ki Ghamti " button par ek baar click karein. Iske baad audio unblock ho jayegi.

Issue / Question	Underlying Cause	Step-by-Step Resolution
Rider ko customer ka ghar nahi mil raha?	Pahaadon par house number nahi hote, bas landmarks aur seedhiyan hoti hain.	Rider app me Staircase Guidance Note padhein ya seedhe "📞 Call Customer " button dabayein. Number masked rehta hai.
Customer OTP bhool gaya ya phone switch off ho gaya?	Battery drain in freezing mountain cold.	Rider Merchant ko call karke Admin se Master Bypass OTP request kar sakta hai, jiska record audit logs me save hota hai.
Thermal Slip ka print cut ho raha hai?	Printer paper roll width mismatch (58mm vs 80mm).	Thermal Slip Preview modal me paper width 58mm / 80mm toggle check karein. Chrome Print dialog me margins " <i>None</i> " set karein.
Offline mode me order queue flush nahi hua?	Network reconnection event browser ne register nahi kiya.	Customer App me header par " Tunnel Mode " button ko ek baar off-on karein, pending orders turant sync ho jayenge.

PAHADICART • OFFICIAL COMPLETE OPERATIONS BLUEPRINT & USER HANDBOOK

Engineered for the Steep Hills, Ancient Bazaars, and Resilient Communities of Himachal Pradesh • Version 2.0 (2026)

Live Cloud Production: <https://pahadicart.vercel.app>